Theorem 14: upper bound for the spectral distance

Proof overview of the Lean 4/Mathlib formalisation

Folder `GsimOT/Theorem15_SpectralUpperBound`

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport.*
File `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository; statement, equation, page and line numbers refer to this PDF. `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

1 Statement

Paper. Theorem 14 (Section 3.4, p. 12, l. 397–398): *for all graphs G, H of the same order,* $\delta_2(G,H) \le \delta_2^{\mathrm{OT}}(G,H) + 2\sqrt{2\,\delta_2^{\mathrm{OT}}(G,H)}$. The text after it (l. 399) derives $\delta_2 \le (1+2\sqrt{2})\sqrt{\delta_2^{\mathrm{OT}}}$ from $\delta_2^{\mathrm{OT}} \le 1$; this is the δ_2 bound of Theorem 2 (p. 3, l. 119). The proof is App. F.2 (pp. 46–50), with Eqs. (40)–(43) and Lemmas 50–52. It ends (p. 50, l. 1573–1575) with the sharper bound $\delta_2 \le \gamma_Q + 4\sqrt{\gamma_Q h}$, $\gamma_Q = \delta_2^{\mathrm{OT}}$, $h = \min\{\sqrt{p_G(1-p_G)}, \sqrt{p_H(1-p_H)}\}$, $p_G = 2|E(G)|/n^2$, and $h \le \frac{1}{2}$.

Lean. For finite types V, W with $|V| = |W|$ and simple graphs G on V, H on W, writing $\delta = \delta_2^{\mathrm{OT}}(G,H)$ (`deltaOTPQ 2 2 G H`) and $\delta_2 =$ `deltaPQ 2 2 G H`:

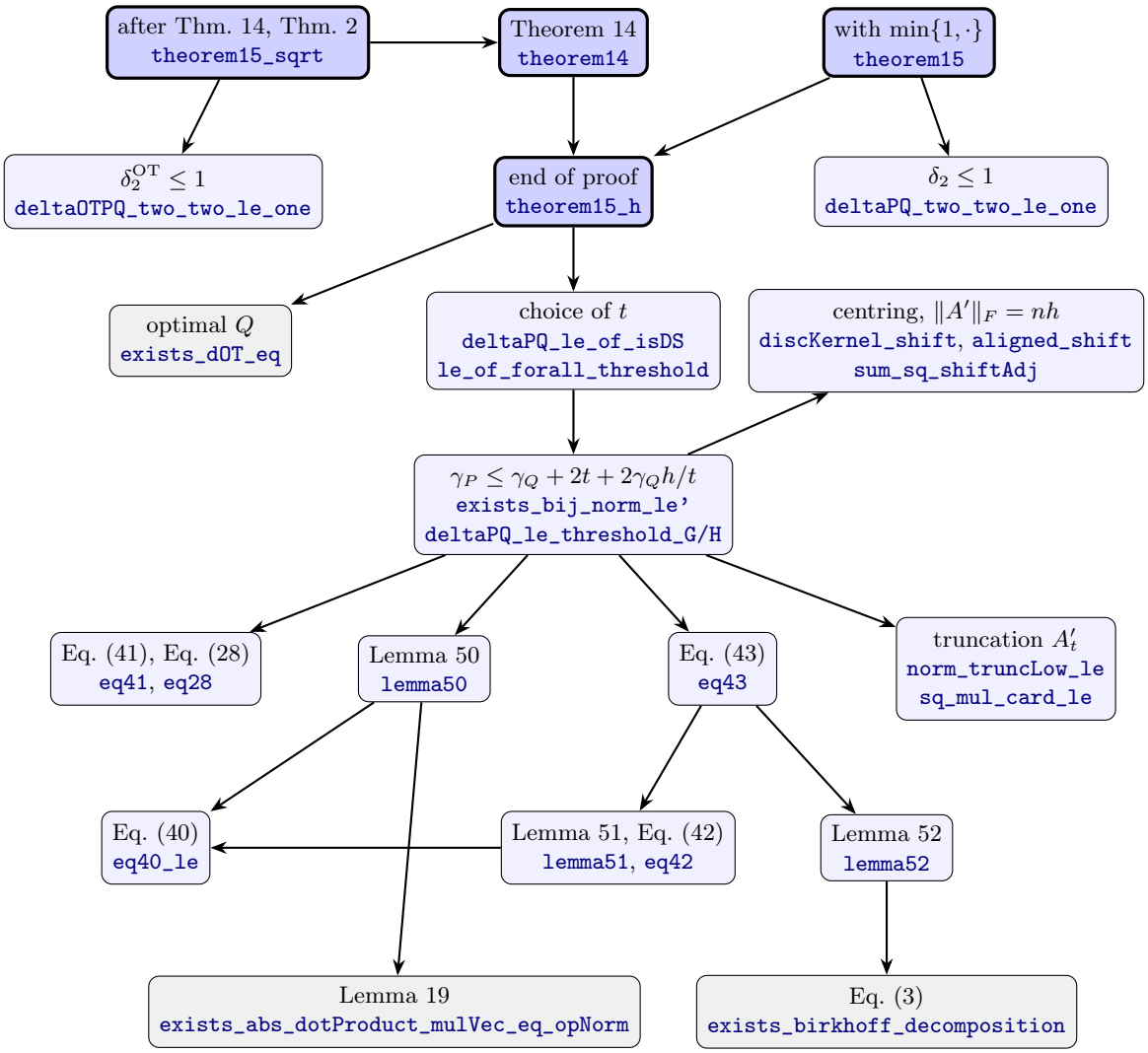
- `GsimOT.theorem14`: $\delta_2(G,H) \le \delta + 2\sqrt{2\delta}$ (exactly Theorem 14);
- `GsimOT.theorem15_h`: $\delta_2(G,H) \le \delta + 4\sqrt{h\,\delta}$ (the end of the proof);
- `GsimOT.theorem15`: $\delta_2 \le \min\{1, \delta + 4\sqrt{h\delta}\} \le \min\{1, \delta + 2\sqrt{2\delta}\}$ (`theorem15_h` combined with $\delta_2 \le 1$, `deltaPQ_two_two_le_one`; it implies Theorem 14);
- `GsimOT.theorem15_sqrt`: $\delta_2(G,H) \le (1+2\sqrt{2})\sqrt{\delta}$ (l. 399 and Theorem 2).

h is written `min (Real.sqrt (edgeDensity G * (1 - edgeDensity G))) (Real.sqrt (edgeDensity H * (1 - edgeDensity H)))` with `edgeDensity` $= p_G$ (`edgeDensity_eq`: $p_G = 2|E(G)|/n^2$). Nothing is cited.

2 Where the proof lives

File	Content
`Tools.lean`	the spectral norm: `opNorm_two_two_eq` ($\|\cdot\|_{2\to2}$ is Mathlib's ℓ^2 operator norm), `l2_norm_le_frobenius`, `l2_norm_le_one_of_stochastic` ($\|R\|_2 \le 1$), transposition and reindexing, the variance identity `sum_weighted_sq_sub`.
`RandomBijection.lean`	the matrix part of App. F.2: the kernel `discKernel` ($n\gamma_Q$), Eq. (40) (`eq40_le`), Lemma 50, Eq. (42), Lemma 51, Lemma 52, Eqs. (41), (28), the spectral truncation, Eq. (43), the chain (a), and the choice of t.
`Theorem15.lean`	the graph part: centring by p_G, the optimal Q, the normalised estimate, and the main theorems.
`Common/Attainment.lean`	`exists_dOT_eq`: an optimal $Q \in \mathrm{DS}(V,W)$ exists.
`Common/BirkhoffDecomposition.lean`	Eq. (3): `exists_birkhoff_decomposition`.
`Common/OperatorNorm.lean`	Lemma 19: `exists_abs_dotProduct_mulVec_eq_opNorm`.
`Common/OTDistances.lean`	$B_{2\to2}$ (`Bmat`), d_2^{OT}, δ_2^{OT}, `deltaOTPQ_eq_dOT`.

3 Proof layout



Arrows mean "uses". Blue: this folder; grey: `GsimOT/Common`. Nothing is cited.

4 The proof step by step

Matrices are indexed by V and W rather than $[n]$; the Lean works with the unnormalised $\|B_{2\to2}(G,H,Q)\|_2 = n\gamma_Q$ and the threshold $s = nt$, and all statements become the paper's after dividing by n.

Eq. (40) (p. 47, l. 1463–1470). $\gamma_Q(A',B')$ is the maximum of $\frac{1}{n^2}|\sum f(i,j)Q_{ij}Q_{k\ell}(A'_{ik} - B'_{j\ell})g(k,\ell)|$ over $\|f\|_Q = \|g\|_Q = 1$. Substituting $\sqrt{Q}f$ for f, this is n^{-1} times the spectral norm of the kernel $\sqrt{Q(a)}(A'_{a_1b_1} - B'_{a_2b_2})\sqrt{Q(b)}$ (`discKernel`); for adjacency matrices this kernel is $B_{2\to2}(G,H,Q)$, also after centring (`discKernel_shift`). The bilinear bound used twice below is `eq40_le`: the sum is $\langle\sqrt{Q}f, K\sqrt{Q}g\rangle$, bounded by Cauchy–Schwarz.

Eq. (41) (l. 1471–1475). $A - PBP^\top = (A - RAR^\top) + P(Q^\top AQ - B)P^\top$ with $R = PQ^\top$: `eq41` (pure algebra; `pmat` is P_π).

Lemma 50 (l. 1478–1483). $n^{-1}\|Q^\top A'Q - B'\|_2 \le \gamma_Q(A',B')$: `lemma50`. As in the paper, Lemma 19 (`exists_abs_dotProduct_mulVec_eq_opNorm`) gives unit vectors x, y attaining the norm, and Eq. (40) is tested with $f(i,j) = y_j$, $g(k,\ell) = x_\ell$; since the columns of Q sum to one the expression is $y^\top(Q^\top A'Q - B')x$ and the lifts have norm ≤ 1 (`sum_Q_mul_snd`). $\|R\|_2 \le 1$ for $R \in \mathrm{DS}(n)$ (l. 1485–1486) is `norm_pmat_mul_transpose_le`, via `l2_norm_le_one_of_stochastic`.

Lemma 51 and Eq. (42) (p. 48, l. 1492–1509). With $F_{ij} = \sum_{k\ell} Q_{k\ell}(A'_{ik} - B'_{j\ell})y_k$ (`Fmat`) $= x_i - r_j$ (`Fmat_eq`), Eq. (42) $\sum Q_{ij}F_{ij}^2 \le (n\gamma_Q)^2\|y\|^2$ is `eq42`: $S^2 = \sum QF^2$ is the expression of Eq. (40) for $f = F$, $g(k,\ell) = y_k$, so $S^2 \le n\gamma_Q S\|y\|$, and $S = 0$ is trivial. Lemma 51 (`lemma51`): $(Q^\top x)_j = r_j + m_j$, $x_i - (Q^\top x)_j = F_{ij} - m_j$, and the expansion $\sum_i Q_{ij}(F_{ij} - m_j)^2 = \sum_i Q_{ij}F_{ij}^2 - m_j^2$ is the variance identity `sum_weighted_sq_sub` with $c = 0$.

Lemma 52 (pp. 48–49, l. 1518–1522). The random permutation P with $\mathbb{E}[P] = Q$ is the one of a Birkhoff decomposition $Q = \sum_\sigma \lambda_\sigma P_{\pi_\sigma}$ (`exists_birkhoff_decomposition`); expectations are the sums $\sum_\sigma \lambda_\sigma(\cdot)$. `lemma52`: $\sum_\sigma \lambda_\sigma\|(I - R_\sigma)x\|^2 = \sum_{ij} Q_{ij}(x_i - (Q^\top x)_j)^2$, via $((I - R)x)_i = x_i - (Q^\top x)_{\pi(i)}$ (`one_sub_R_mulVec`) and linearity (`sum_birkhoff_eq`).

Optimal Q and centring (p. 49, l. 1526–1532). The proof starts from an optimal Q: `exists_dOT_eq` (App. A), and $\gamma_Q = \delta_2^{\mathrm{OT}}$ by `deltaOTPQ_eq_dOT` (in `theorem15_h`). Centring $A' = A - p_G J$, $B' = B - p_G J$ (`shiftAdj`) changes neither D (`discKernel_shift`) nor $A^\pi - B$ (`aligned_shift`), and $\|A'\|_F^2 = n^2(p_G - p_G^2) = n^2h^2$ (`sum_sq_shiftAdj`, using $A_{ik}^2 = A_{ik}$). The paper assumes $h = \sqrt{p_G(1-p_G)}$ "by symmetry" (l. 1529); the Lean proves both cases (`deltaPQ_le_threshold_G`, `deltaPQ_le_threshold_H`; the second centres by p_H and applies the argument to $(B', A', Q^\top)$, `exists_aligned_le'`).

Spectral truncation (p. 49, l. 1533–1544). $A' = \sum_r \lambda_r\phi_r\phi_r^\top$ (`eigU_spectral`), $A'_t = \sum_{r\in I_t} \lambda_r\phi_r\phi_r^\top$ (`truncHigh`, with the indicator `ind` of $I_t = \{r : |\lambda_r| > s\}$, $s = nt$), $\|A' - A'_t\|_2 \le nt$ (`norm_truncLow_le`) and $|I_t|\,(nt)^2 \le \|A'\|_F^2$ (`sq_mul_card_le`, via $\sum_r \lambda_r^2 = \|A'\|_F^2$, `sum_sq_entries_eq`).

Eq. (43) (pp. 49–50, l. 1552–1562). `eq43`: $\mathbb{E}\|EA'_t\|_2^2 \le \mathbb{E}\|EA'_t\|_F^2 = \sum_{r\in I_t}\mathbb{E}\|EA'\phi_r\|_2^2 \le (n\gamma_Q)^2|I_t|$. The first step is $\|\cdot\|_2 \le \|\cdot\|_F$; the equality is $\|K\|_F^2 = \sum_r \|K\phi_r\|^2$ (`frobenius_eq_sum_eigvec`; l. 1453 and l. 1554–1556) together with $A'_t\phi_r = A'\phi_r$ for $r \in I_t$ and 0 otherwise (`truncHigh_mulVec_eigvec`, `frobenius_one_sub_R_truncHigh`); then Lemma 52 and Lemma 51 with $x = A'\phi_r$, $y = \phi_r$, $\|\phi_r\| = 1$. Some outcome is at most the expectation (`exists_le_of_sum_mul_le`), which gives a permutation with $\|EA'_t\|_2 \le n\gamma_Q h/t$ (`exists_one_sub_R_truncHigh_le`).

Chain (a) and the main estimate (p. 50, l. 1563–1572). `exists_bij_norm_le'` splits $A' - RA'R^\top = ((A' - A'_t) - R(A' - A'_t)R^\top) + (A'_t - RA'_tR^\top)$, applies Eq. (28) to A'_t (`eq28`: $A'_t - RA'_tR^\top = EA'_t + R(EA'_t)^\top$), uses $\|R\|_2 \le 1$, and obtains $\|A' - RA'R^\top\|_2 \le 2nt + 2n\gamma_Q h/t$; with Eq. (41) and Lemma 50, $\|A' - PB'P^\top\|_2 \le n\gamma_Q + 2nt + 2n\gamma_Q h/t$. `exists_bij_norm_le` transfers this to $P^\top A'P - B'$ (the orientation $A_G^\pi - A_H$ of d_2), and `deltaPQ_le_threshold_G` divides by n: $\delta_2 \le \gamma_P \le \gamma_Q + 2t + 2\gamma_Q h/t$.

Choice of t (l. 1573–1575). `le_of_forall_threshold`: $t = \sqrt{\gamma_Q h}$ if $\gamma_Q h > 0$, otherwise $t \to 0$; this gives `deltaPQ_le_of_isDS` and `theorem15_h`. Finally $h \le \frac{1}{2}$ (`min_sqrt_density_le_half`) gives $4\sqrt{h\gamma_Q} \le 2\sqrt{2\gamma_Q}$ (`four_sqrt_le`) and `theorem14`.

After Theorem 14 (p. 12, l. 399). $\delta_2^{\mathrm{OT}} \le 1$ is `deltaOTPQ_two_two_le_one`: for every coupling Q^*, $\|B_{2\to2}(G, H, Q^*)\|_2 \le \|\cdot\|_F \le \sum Q^* \le 1$ (`opNorm_Bmat_le_sum`). Then $\delta \le \sqrt{\delta}$ and `theorem15_sqrt` follows from `theorem14`.

5 Assumptions and trusted base

No cited result is used and there are no hypotheses beyond $|V| = |W|$. The axioms are `propext`, `Classical.choice`, `Quot.sound`. Shared results: Eq. (3) (derived from Mathlib's Birkhoff–von Neumann theorem), Lemma 19, attainment of d_2^{OT}, the spectral theorem of Mathlib.

6 Notes on the formalisation

- **Statement.** `theorem14` is the paper's statement; `theorem15` combines the bound `theorem15_h` at the end of the proof with $\delta_2 \le 1$ (`deltaPQ_two_two_le_one`) into $\delta_2 \le \min\{1, \delta + 4\sqrt{h\delta}\} \le \min\{1, \delta + 2\sqrt{2\delta}\}$.
- **No symmetry reduction** for h: both cases are proved.

- **Normalisation and randomness:** $n\gamma_Q$ instead of γ_Q, threshold $s = nt$; expectations are sums over a Birkhoff decomposition.

7 How to check this result

```
lake build GsimOT.Theorem15_SpectralUpperBound.Theorem15
lake env lean scripts/CheckAxioms.lean   # prints the axioms of GsimOT.theorem15, theorem15_sqrt
```

For a single theorem, add `#print axioms GsimOT.theorem14` to a file importing `GsimOT.Theorem15_SpectralUpperBound`. The table in `Theorem15_correspondence.pdf` lists every step with file and line.